Class | Adm No

Candidate Name: ______________________________

H

2013 Promotional Examination II
Pre-University 2

H1 Biology **Syllabus 8875/01**

12 September 2012

1 hour

Additional material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

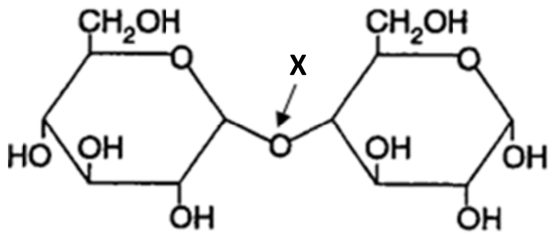
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

Answer **all** questions.

1. The diagram below shows a disaccharide.



What is the bond labelled **X**?

A. α (1→4) glycosidic bond

B. α (1→6) glycosidic bond

C. α (1→4) peptide bond

D. α (1→6) peptide bond

2. A student discovered a new type of seed in his back garden. He crushed the germinating seeds and tested the seed extract. The table below shows the results of his tests.

Food test	Result
Benedict's test	Positive – brick red precipitate observed
Biuret test	Positive – purple colour observed
Fat emulsion test	Remained clear
Iodine test	Remained yellow-brown

Which molecules are present in the seed extract?

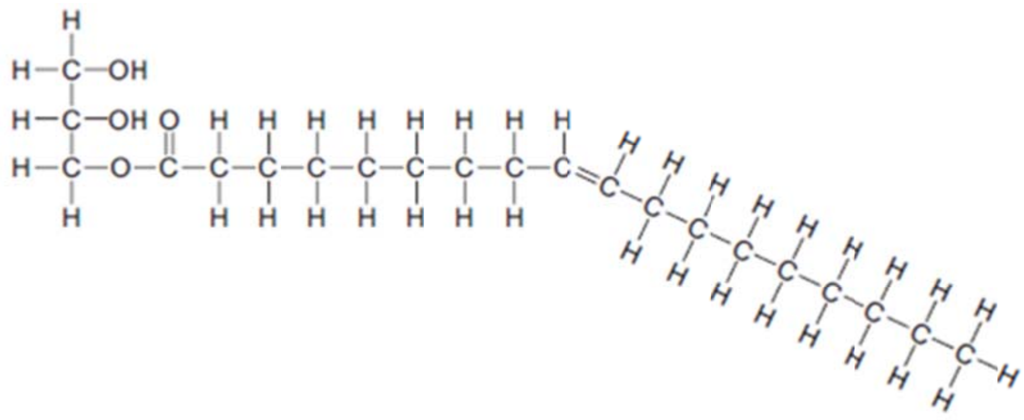
A. fructose and protein only

B. maltose and protein only

C. maltose, fat and protein only

D. fructose, fat and starch only

3. The diagram shows a triglyceride molecule that has been partially hydrolysed.



If the above molecule were to be completely hydrolysed, what would be the products formed?

A. one molecule of glycerol and one molecule of saturated fatty acid

B. one molecule of glycerol and one molecule of unsaturated fatty acid

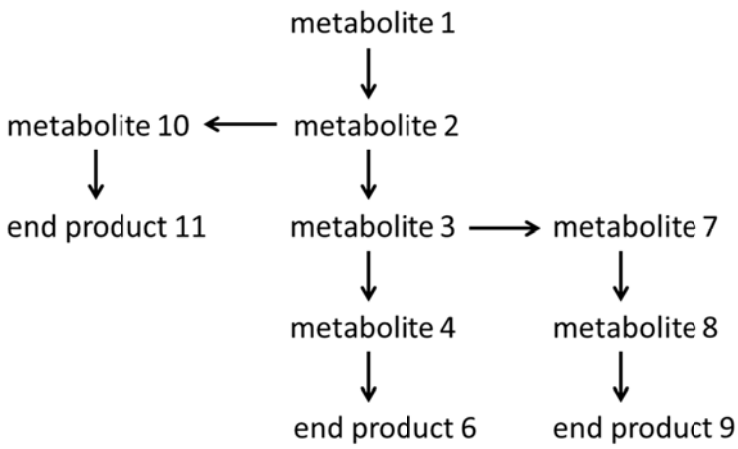
C. one molecule of glycerol, one molecule of unsaturated fatty acid and one molecule of water

D. one molecule of glycerol, one molecule of saturated fatty acid and one molecule of water

4. The table below shows the different types of bonds that contribute to a protein's 3D shape. Which of the following options correctly matches the bond type to the bond strength?

	Peptide bond	**Ionic bond**	**Hydrogen bond**	**Disulphide bridge**
A.	Weak	Weak	Strong	Weak
B.	Strong	Strong	Weak	Strong
C.	Strong	Weak	Weak	Strong
D.	Weak	Strong	Strong	Weak

5. The diagram below shows a metabolic pathway that is controlled by end product inhibition. When the concentration of the end products increase beyond a particular level, an enzyme in the pathway is inhibited to lower the rate of production.



It was discovered that an increase in amount of end product 9 does not affect the synthesis of end product 11, however, it decreases synthesis of end product 6.

The enzyme controlling which of the following reactions could be inhibited by end product 9?

A. metabolite 1 → metabolite 2

B. metabolite 2 → metabolite 3

C. metabolite 3 → metabolite 7

D. metabolite 2 → metabolite 10

6. Which of the following statements correctly describes non-competitive inhibition?

   A. An increase in substrate concentration will decrease the effect of inhibition on the rate of reaction.

   B. The inhibitor binds at the active site and competes with the substrate for the active site.

   C. As substrate concentration increases, the maximum rate of reaction will always be lower than a situation where no inhibitor is added.

   D. The inhibitor forms a reversible bond with a site other than the active site and causes a change in shape of the active site.

7. Which of the following features of mitochondria and chloroplasts provides evidence for the endosymbiotic theory?

   A. Both have circular DNA.

   B. Both are able to synthesise ATP.

   C. Both are able to survive independently outside of the eukaryotic cell.

   D. Both have 80S ribosomes.

8. An unknown chemical X was found to be able to inhibit the process of mitosis if it was added to cells between metaphase and anaphase. However, chemical X had no effect on mitosis if it was added after anaphase.

   What could be a possible direct effect of chemical X?

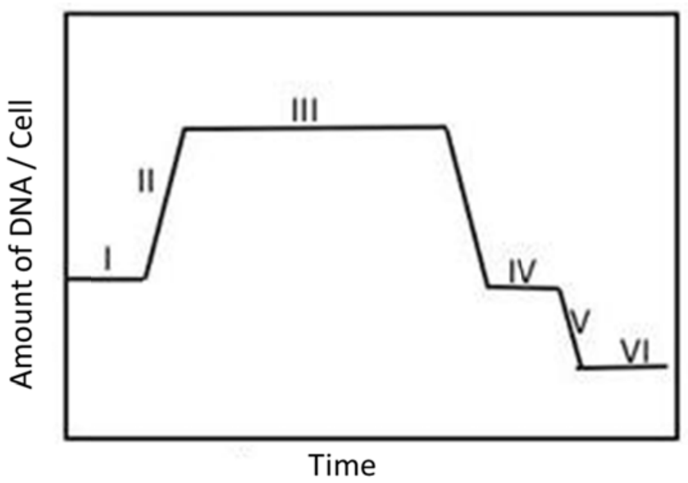
   A. Inhibition of microtubule attachment to kinetochores of chromosomes.

   B. Increased rate of reformation of the nuclear envelope.

   C. Inhibition of the condensation of chromosomes.

   D. Increased rate of shortening of microtubules in anaphase.

9. The graph below shows the change in DNA content of a cell throughout the process of meiosis.



With reference to the graph, which of following statements is correct?

A. DNA replication is occurring at phase I.

B. Homologous chromosomes move to opposite poles of the cell at phase II.

C. The cell contains haploid number of chromosomes at phase IV.

D. Each cell contains pairs of homologous chromosomes at phase VI.

10. A mutation in a gene was observed to have no effect on the protein that is formed.

Which of the following is least likely to account for this observation?

A. The mutation was a single base substitution in the third nucleotide of a codon.

B. The mutation was a single base deletion in the third nucleotide of a codon.

C. The mutation was a three base deletion in the exon of the gene.

D. The mutation was a three base substitution in the intron of the gene.

11. Which of the following helps to ensure the integrity of DNA during DNA replication?

A. phosphodiester bonds between adjacent nucleotides

B. covalent bonds between the deoxyribose sugar and the nitrogenous base

C. hydrophobic interactions between neighbouring DNA molecules

D. hydrogen bonds between nitrogenous bases on opposite DNA strands

12. Scientists have invented an anti-sense genetic drug that aims to lower expression levels of a particular protein. This anti-sense drug is made of a short sequence of RNA nucleotides that are complementary to the protein-coding mRNA.

Which of the following suggests a possible mode of action of this anti-sense drug?

A. It binds to the anti-sense strand of DNA and prevents replication of that particular gene.

B. It binds to the sense strand of DNA and prevents transcription of the gene.

C. It binds to the protein-coding mRNA and prevents translation of the protein.

D. It binds to the newly synthesized protein and prevents functioning of the protein.

13. Which of the following is most likely to result in cancer?

A. Inactivation of a single tumour suppressor gene.

B. Inactivation of both a proto-oncogene and a tumour-suppressor gene.

C. Hyperactivity of a proto-oncogene and inactivation of a tumour suppressor gene.

D. Hyperactivity of both a proto-oncogene and a tumour suppressor gene.

14. The table below shows the mRNA codons for a number of amino acids.

AAU	Asparagine	AGC	Serine	ACC	Threonine
CCA	Proline	CCU	Proline	CGG	Arginine
UCC	Serine	GGA	Glycine	UGC	Cysteine

The DNA sequence GGATGGGCCGGT codes for a tetrapeptide. However, a mutagen results in thymine in DNA pairing with uracil in RNA instead.

Which of the following tetrapeptides could be synthesised from the DNA sequence as a result of the mutagen?

A. proline – threonine – arginine - proline

B. glycine – threonine – arginine – glycine

C. glycine – serine – arginine - proline

D. proline – serine – arginine – proline

15. The table below shows some diseases and their causes.

	disease	**cause**
1	brittle bone disease	abnormal collagen protein
2	Down's syndrome	extra chromosome 21
3	sickle cell anaemia	abnormal haemoglobin
4	Turner's syndrome	absence of one sex chromosome
5	Cri-du-chat syndrome	missing section from chromosome 5

Which diseases are caused by gene mutation and which by chromosome mutation?

	gene mutation	**chromosome mutation**
A.	**1** and **3**	**2** and **5**
B.	**1** and **5**	**2** and **4**
C.	**3** and **5**	**2** and **4**
D.	**3** and **5**	**1** and **2**

16. In an investigation to determine whether two characteristics, eye colour and wing shape, of Drosophila flies were located on autosomal chromosomes or on sex chromosomes, two crosses were carried out:

Cross 1	red eye and normal wing female x white eye and vestigial wing male
Cross 2	white eye and vestigial wing female x red eye and normal wing male

In cross 1, all offspring had red eyes and normal wings. However, in cross 2, although all offspring had normal wings, all females had red eyes while all males had white eyes.

Based on the results of the two crosses, which of the following statements correctly describes inheritance of these two characteristics?

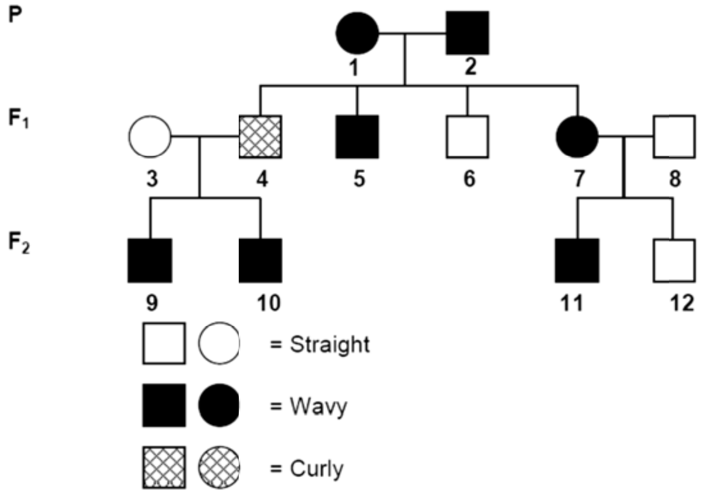
A. Both genes are sex-linked.

B. Both genes are autosomally inherited.

C. Eye colour is sex-linked while wing shape is autosomally inherited.

D. Wing shape is sex-linked while eye colour is autosomally inherited.

17. The pedigree below shows the inheritance of hair type in a family.



If person 10 marries an individual who has curly hair, what is the probability that their first child will be a daughter with curly hair?

A. 0%

B. 25%

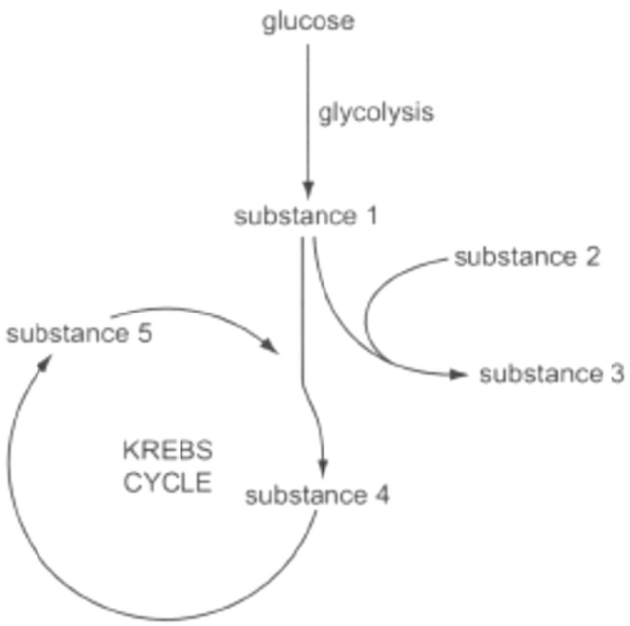
C. 50%

D. 100%

18. Which combination correctly describes the light-independent reactions of photosynthesis?

	site of reaction	ATP	NADP
A.	stroma	ATP → ADP + P_i	oxidised
B.	thylakoid membrane	ATP → ADP + P_i	reduced
C.	stroma	ADP + P_i → ATP	oxidised
D.	thylakoid membrane	ADP + P_i → ATP	reduced

19. The diagram below shows some stages of aerobic and anaerobic respiration in animal cells.



Which of the following correctly identifies substances 1, 2, 3, 4 and 5?

	1	2	3	4	5
A.	acetyl coA	NAD	ethanol	pyruvate	6 carbon compound
B.	acetyl coA	reduced NAD	lactate	6 carbon compound	4 carbon compound
C.	pyruvate	NAD	ethanol	acetyl coA	6 carbon compound
D.	pyruvate	reduced NAD	lactate	6 carbon compound	4 carbon compound

20. Tsetse flies carry diseases that can infect zebras. It has been observed that animals with horizontal stripes are bitten less frequently by tsetse flies.

Based on these observations, which of the following could be true?

A. Tsetse flies act as a selection pressure. Zebras with more horizontal stripes survive longer and breed more, passing the allele for more stripes on to their offspring.

B. Bites from tsetse flies can cause mutations in the zebra genome. Zebras with mutations die before they can pass on their genes to their offspring, thus zebras with fewer stripes have fewer offspring.

C. Tsetse flies act as a selection pressure. Zebras gradually develop more horizontal stripes in their lifetime in order to increase resistance to tsetse flies, this trait is passed on to their offspring.

D. Bites from tsetse flies can cause mutations in the zebra genome. These mutations cause zebras to gradually develop fewer stripes and become more susceptible to the diseases caused by the flies.

21. Which of the following is a pair of analogous structures?

A. human limb and dolphin flippers

B. human cytochrome C and chimpanzee cytochrome C

C. fish circulatory system and pig circulatory system

D. bat wings and fly wings

22. Two species of frogs belonging to the same genus occasionally mate successfully, however, their offspring fail to develop to maturity, as a result they are still considered separate species.

What mechanism keeps these two species of frogs separate?

A. postzygotic barrier known as hybrid sterility

B. postzygotic barrier known as hybrid inviability

C. prezygotic barrier known as gametic barrier

D. prezygotic barrier known as behavioural isolation

23. Which of the following is not a mechanism that can shift allele frequencies in populations?

A. stabilising selection

B. bottleneck effect

C. gene flow between neighbouring populations

D. acquiring of new traits in an organism's lifetime

24. Reporter genes such as the *lacZ* gene and ampicillin resistance gene are often included in plasmid vectors.

What is the purpose of having reporter genes in recombinant plasmids?

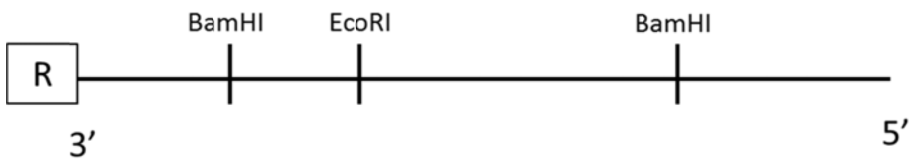
A. facilitates entry of the recombinant plasmid into the bacterial host cell

B. provides a mechanism for identifying or quantifying gene expression

C. allows flexible switching between prokaryotic to eukaryotic host cells

D. provides a means to identify suitable host cells

25. The diagram below shows a restriction map of a linear DNA sequence. This linear sequence has been radioactively labelled (R) at the 3' end of the DNA.



This linear DNA undergoes complete digestion using the restriction enzyme BamHI.

Which of the following correctly summarises the different radioactively labelled and non-labelled restriction fragments that can be found in the final digestion mix?

A. 1 labelled and 1 non-labelled

B. 1 labelled and 2 non-labelled

C. 2 labelled and 2 non-labelled

D. 2 labelled and 3 non-labelled

26. Which of the following is not a potential use for the polymerase chain reaction?

A. Detection of mutations associated with cancer.

B. Amplification of DNA from unusual sources such as fossilised organisms.

C. Cloning of genes for genetic manipulation.

D. DNA fingerprinting for crime scene investigations.

27. In order to mass produce human insulin, genetic engineering of bacteria has to be carried out whereby the human insulin gene is introduced into bacterial cells. The following steps are required for the genetic engineering process:

1	Ligation of human insulin gene with plasmid vector using DNA ligase
2	Identifying bacteria that have taken up recombinant plasmid via replica plating
3	Transformation of bacteria through heat shock
4	Scaling up production by growing bacterial colonies in a bioreactor
5	Digestion of human insulin gene and bacterial plasmid with restriction enzymes
6	Identifying bacterial colonies containing human insulin gene using gene probe

Which of the following shows the correct order of steps in order to genetically engineer bacteria with the human insulin gene?

A. **1 → 5 → 4 → 3 → 2 → 6**

B. **1 → 5 → 3 → 2 → 6 → 4**

C. **5 → 1 → 4 → 3 → 2 → 6**

D. **5 → 1 → 3 → 2 → 6 → 4**

28. The cloning of a eukaryotic gene in a prokaryotic host cell involves a number of difficulties, one of which can be solved with the use of mRNA and reverse transcriptase.

Which is the difficulty that can be overcome by this solution?

A. presence of introns and exons in eukaryotic mRNA

B. presence of 5' capping in eukaryotic mRNA

C. simultaneous transcription and translation in prokaryotes

D. lack of RNA polymerase in prokaryotes

29. Which of the following are similarities between all cancer cells and all stem cells?

A. They replicate indefinitely and are able to move from one location in the body to another.

B. They lack cell-cell adhesion and are non-differentiated.

C. They replicate indefinitely and are found in various parts of the body.

D. They are able to move from one location in the body to another and are non-differentiated.

30. What are the normal functions of stem cells in the body of a human?

A.	Differentiating into many kinds of cells in a 3 – 5 day old human embryo	Producing insulin in pancreases damaged by type 1 diabetes	Cells that can differentiate into bone cells in the skeletons
B.	Differentiating into many kinds of cells in a 3 – 5 day old human embryo	Producing red blood cells worn out by normal wear and tear	Cells that can differentiate into cardiac muscle cells in the heart
C.	Differentiating into only one kind of cells in a 3 –5 day old human embryo	Producing insulin in pancreases damaged by type 1 diabetes	Cells that can differentiate into cartilage cells in the joints
D.	Differentiating into only one kind of cells in a 3 – 5 day old human embryo	Producing red blood cells worn out by normal wear and tear	Cells that can differentiate into nerve cells in the brain